

Introduction

In this SGTA, you will continue to analyse the dataset introduced in week 7 (**health.csv**, available on iLearn). As a reminder, *this study investigates the effects of different diet and exercise interventions on a variety of physiological, behavioural, and self-reported health outcomes in a diverse adult sample. A total of 1,500 participants were recruited from four urban gyms and randomly assigned to combinations of diet plans (Low Carb, Mediterranean, Intermittent Fasting) and exercise regimens (Cardio, Strength, Mixed). Data collected include anthropometric measures (e.g., weight loss), cardiovascular indicators (e.g., resting heart rate), metabolic markers (e.g., cholesterol), psychological and behavioural measures (e.g., reaction time, satisfaction). The aim is to assess the effects of interventions, test group differences, and evaluate key assumptions of classical statistical methods in a realistic setting.*

The variables in the dataset are listed below:

- **id**: participant identifier
- **gender**: Male/Female. Categorical
- **age**: age in years. Integer
- **gym**: one of East, South, West, North. Categorical
- **diet**: one of Mediterranean, Intermittent Fasting, Low Carb. Categorical
- **exercise**: one of Cardio, Mixed, Strength. Categorical
- **fitness_score_before**: fitness score (0-100) at enrolment in the study. Integer
- **fitness_score_after**: fitness score (0-100) at the end of the study. Integer
- **weight_loss**: weight loss in kg. Numerical continuous.
- **cholesterol_before**: cholesterol level at enrolment in the study. Integer
- **cholesterol_after**: cholesterol level at the end of the study. Integer
- **met_goal**: if the goal (5kg weight loss) has been met. 0/1
- **success**: if the fitness score has improved by 5. 0/1
- **satisfaction**: the level of satisfaction (1-5). Categorical ordinal
- **week1**: the weight loss in week 1. Numerical continuous
- **week2**: the weight loss in week 2. Numerical continuous
- **week3**: the weight loss in week 3. Numerical continuous
- **week4**: the weight loss in week 4. Numerical continuous
- **rhr_before**: Resting Heart Rate (RHR) at enrolment in the study. Numerical continuous
- **rhr_after**: Resting Heart Rate (RHR) at the end of the study. Numerical continuous
- **reaction_time_before**: Reaction time (RT) in milliseconds at enrolment in the study. Numerical continuous
- **reaction_time_after**: Reaction time (RT) in milliseconds at the end of the study. Numerical continuous

Expand the report started in week 7 and address the following task.

1. EDA (Exploratory Data Analysis). Use numerical and graphical tools to illustrate the data, including single-variable summaries and relations among variables. Focus your efforts on the variables considered in the questions below.

2. Answer the following questions:

- Compute the difference in fitness score (`fitness_score_diff`) between enrolment and end of study:
 - Which exercise plan differ in `fitness_score_diff`? *Tukey's HSD*
 - Does Strenght exercise have a different effect than the average of Cardio and Mixed? *contrast*
- Is there a difference in satisfaction between males and females? *Mann-Whitney U test*
- Is the mean weight loss different between males and females? *Kruskal-Wallis test*
- Does weight loss vary by diet plan? *Kruskal-Wallis test*